# Computer Architecture
# CS 622 / EE 520
# Fall 2026

Lecture 2

Shahid Masud

# Topics

- Look at 3 Papers for upcoming Computer Architectures
- Evolution in Computing in post Moore's Law era
- RISC-V Instruction Set Architecture
- Introducing Computer Performance

# Parallelism in Computer Architecture

# Amdahl's Law

*Amdahl's Law* (Section 1.9) prescribes practical limits to the number of useful cores per chip. If 10% of the task is serial, then the maximum performance benefit from parallelism is 10 no matter how many cores you put on the chip.

**Some part of code can be processed in parallel**
**The remaining part of code must be processed in a sequential fashion**

# Amdahl's Law – Parallel or Accelerator

*Amdahl's Law* (Section 1.9) prescribes practical limits to the number of useful cores per chip. If 10% of the task is serial, then the maximum performance benefit from parallelism is 10 no matter how many cores you put on the chip.

Amdahl's Law defines the *speedup* that can be gained by using a particular feature. What is speedup? Suppose that we can make an enhancement to a computer that will improve performance when it is used. Speedup is the ratio

$$\text{Speedup} = \frac{\text{Performance for entire task using the enhancement when possible}}{\text{Performance for entire task without using the enhancement}}$$

# Example of Performance Gains through Architecture

**Table 1. Speedups from performance engineering a program that multiplies two 4096-by-4096 matrices.** Each version represents a successive refinement of the original Python code. "Running time" is the running time of the version. "GFLOPS" is the billions of 64-bit floating-point operations per second that the version executes. "Absolute speedup" is time relative to Python, and "relative speedup," which we show with an additional digit of precision, is time relative to the preceding line. "Fraction of peak" is GFLOPS relative to the computer's peak 835 GFLOPS. See Methods for more details.

| Version | Implementation | Running time (s) | GFLOPS | Absolute speedup | Relative speedup | Fraction of peak (%) |
|---|---|---|---|---|---|---|
| 1 | Python | 25,552.48 | 0.005 | 1 | — | 0.00 |
| 2 | Java | 2,372.68 | 0.058 | 11 | 10.8 | 0.01 |
| 3 | C | 542.67 | 0.253 | 47 | 4.4 | 0.03 |
| 4 | Parallel loops | 69.80 | 1.969 | 366 | 7.8 | 0.24 |
| 5 | Parallel divide and conquer | 3.80 | 36.180 | 6,727 | 18.4 | 4.33 |
| 6 | plus vectorization | 1.10 | 124.914 | 23,224 | 3.5 | 14.96 |
| 7 | plus AVX intrinsics | 0.41 | 337.812 | 62,806 | 2.7 | 40.45 |

AVX = Intel Advanced Vector Extension

# Types of Parallelism in Parallel Computers

Parallelism at multiple levels is now the driving force of computer design across all four classes of computers, with energy and cost being the primary constraints. There are basically two kinds of parallelism in applications:

1. *Data-level parallelism (DLP)* arises because there are many data items that can be operated on at the same time.
2. *Task-level parallelism (TLP)* arises because tasks of work are created that can operate independently and largely in parallel.

Computer hardware in turn can exploit these two kinds of application parallelism in four major ways:

1. *Instruction-level parallelism* exploits data-level parallelism at modest levels with compiler help using ideas like pipelining and at medium levels using ideas like speculative execution.
2. *Vector architectures, graphic processor units (GPUs), and multimedia instruction sets* exploit data-level parallelism by applying a single instruction to a collection of data in parallel.
3. *Thread-level parallelism* exploits either data-level parallelism or task-level parallelism in a tightly coupled hardware model that allows for interaction between parallel threads.
4. *Request-level parallelism* exploits parallelism among largely decoupled tasks specified by the programmer or the operating system.

# Seven Great Ideas in Computer Architecture

# Significant Involvement in Computer Architecture

**Explore further**
**In this course**

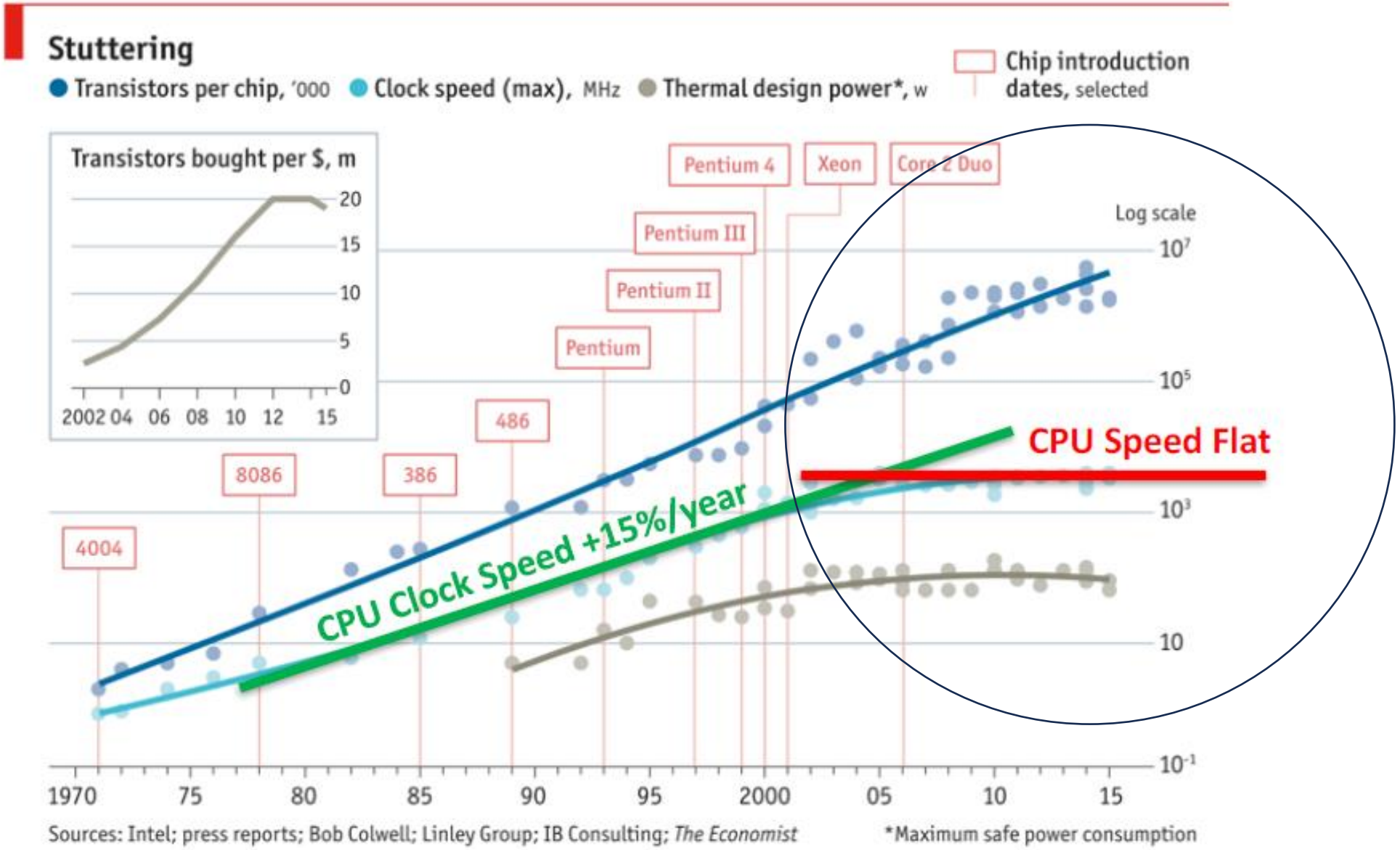
1. Design for ***Moore's Law***
2. Use ***abstraction*** to simplify design
3. Make the ***common case fast***
4. Performance *via* ***parallelism***
5. Performance *via* ***pipelining***
6. Performance *via* ***prediction***
7. ***Hierarchy*** of memories
8. ***Dependability*** *via* redundancy








# Why is Computer Architecture Exciting Today?



# Remaining Lecture

# Interesting Topics

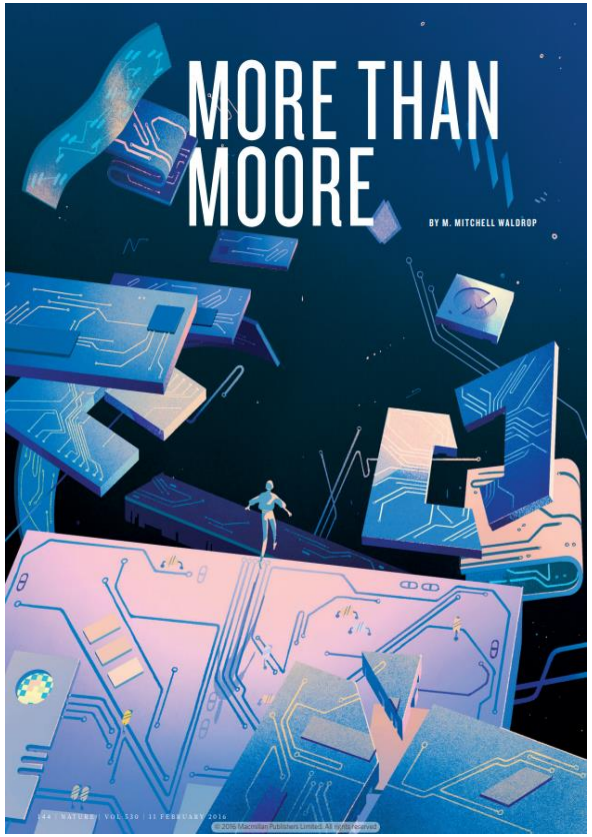
1. Discussion on Moore's Law, through paper 'MORE THAN MOORE' by M. Mitchell Waldrop, published In Nature, February 2016
2. Computer Performance Graphs from National Academy, USA, report
3. Introduction to the specialization area of 'Computer Architecture' through paper by Hennessy and Patterson, 'A New Golden Age for Computer Architecture', published in Communications of the ACM, February 2019, pages 48 to 60.

   **Important Directions for Computers of the future:**

   a. Moore's Law failing to keep up
   b. More Energy dissipation – too hot chips
   c. Domain Specific Architectures
   d. Enhanced security features inside microprocessors
   e. Open Instruction Sets and Extension of Instruction Sets through Customized Accelerators
   f. Agile Hardware Design for Microprocessors
   g. Combination of Domain Specific Languages and Domain Specific Architectures

4. Intel Processor Timeline
5. Reviewed course outline including detailed topics and grading breakup of lectures and labs.

# Look at some recent research papers for inspiration




THEME ARTICLE: MICROPROCESSOR AT 50

## The 50 Year History of the Microprocessor as Five Technology Eras

John L. Hennessy, *Stanford University, Stanford, CA, 94305, USA*

The evolution of the microprocessor can be organized into five eras, each distinguished by common trends in the evolution of microprocessors. Most of these eras are around ten years and represent a shift from the previous era. I have had the privilege of being involved in some way for roughly 48 of the 50 years, so this is also a somewhat personal view.

### FIRST DECADE 1971–1981

This decade, which began with the birth of the Intel 4004 in 1971, was dominated by three trends:

1. an increase in width as Moore's law-enabled processors to go from 4 to 8 to 16 and eventually 32 bits;
2. a rapid increase in instruction sets, often motivated by assembly language examples and enabled by microcode implementations (see the early Intel and Zilog microprocessors);
3. a rapid increase in clock speeds enabled by faster transistors.

The emergence of the personal computer (Apple II in 1977 and IBM PC in 1979) enabled the shrinkwrap software industry and reinforced the importance of object code compatibility, which the first microprocessors did not exhibit. The Motorola 68000, which appeared in production in the late 1980s, was the first 32-bit microprocessor and offered many of the features associated with minicomputers.

### RISC AND PIPELINING YEARS 1981–1995

As Moore's law progressed, it seemed likely that microprocessors would evolve into full blown computers. The accompanying growth in DRAM capacity (from 1 Kib in 1971 to 16 Kib in 1981) reduced the need to hand-optimize code and program in assembly language. The subsequent movement to higher level languages inspired the groups at IBM, Berkeley, and Stanford to explore what became the RISC ideas. In addition to targeting compiler output, rather than handcrafted assembly language, they also emphasized elimination of microcode and compilation to an efficient hardware implementation.

The RISC ideas led to an explosion in the use of pipelining in microprocessors, which generated a rapid increase in clock rate performance. This era was characterized by incredible annual performance growth of approximately 1.5 times, enabled by the inclusion of caches and much faster clocks.

The capstone events in this era were the introduction of 64-bit processors (the R4000, followed by the DEC Alpha, and others) and a growth in pipeline depth from 5 to 7–10 or more stages, which led to clock rates rivaling ECL mainframes.

### INTENSIVE ILP YEARS 1995–2005

The third era is characterized by an intensive focus on exploiting instruction-level parallelism and trying to reduce the clock cycles per instruction to less than 1. The ILP-intensive processors fall into two broad categories: superscalar and very long instruction word (VLIW). The superscalar processors used a combination of hardware techniques and software scheduling to issue more than one instruction per clock, while the VLIW approach relied on little hardware support and intensive compiler scheduling to organize independent operations into issue packets. The VLIW approach did not succeed in the end, due to a variety of factors, most importantly the inability to achieve high performance on less structured integer programs.

The initial superscalar processors (such as the MIPS R8000, PowerPC 604, and later Intel Pentium) used static scheduling, meaning that instruction issue was blocked if the next instruction's operands were not yet available. This approach was rapidly followed by a shift to dynamic scheduling (allowing instructions to be executed out of order) and speculation (allowing instructions to be executed before a preceding branch

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REVIEW SUMMARY

COMPUTER SCIENCE

**There's plenty of room at the Top: What will drive computer performance after Moore's law?**

Charles E. Leiserson, Neil C. Thompson*, Joel S. Emer, Bradley C. Kuszmaul, Butler W. Lampson, Daniel Sanchez, Tao B. Schardl



turing lecture

DOI:10.1145/3282307

**Innovations like domain-specific hardware, enhanced security, open instruction sets, and agile chip development will lead the way.**

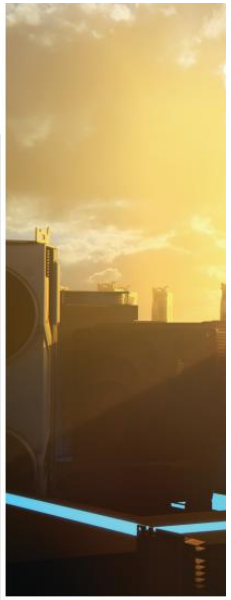
BY JOHN L. HENNESSY AND DAVID A. PATTERSON

# A New Golden Age for Computer Architecture

WE BEGAN OUR Turing Lecture June 4, 2018[1] with a review of computer architecture since the 1960s. In addition to that review, here, we highlight current challenges and identify future opportunities, projecting another golden age for the field of computer architecture in the next decade, much like the 1980s when we did the research that led to our award, delivering gains in cost, energy, and security, as well as performance.

*"Those who cannot remember the past are condemned to repeat it."* —George Santayana, 1905

Software talks to hardware through a vocabulary called an instruction set architecture (ISA). By the early 1960s, IBM had four incompatible lines of computers, each with its own ISA, software stack, I/O system, and market niche—targeting small business, large business, scientific, and real time, respectively. IBM



engineers, including ACM A.M. Turing Award laureate Fred Brooks, Jr., thought they could create a single ISA that would efficiently unify all four of these ISA bases.

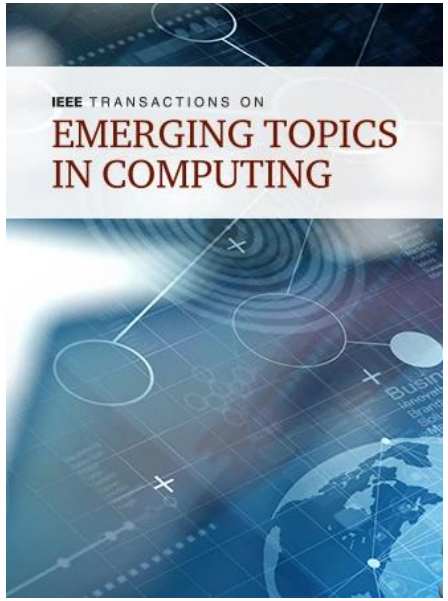
They needed a technical solution for how computers as inexpensive as

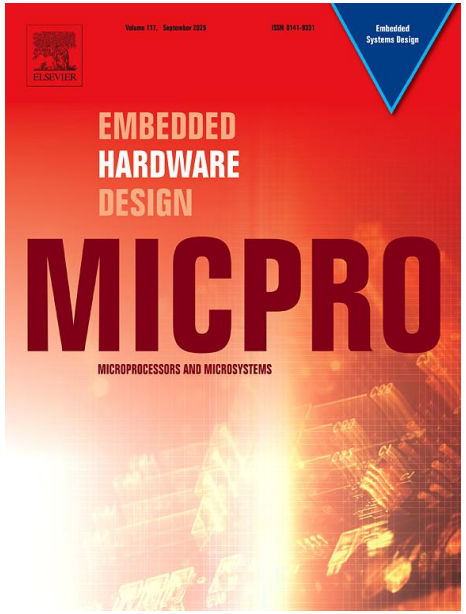
» key insights

- Software advances can inspire architecture innovation.
- Elevating the hardware/software interface creates opportunities for architecture innovation.
- The marketplace ultimately settles architecture debates.

# Some Computer Architecture Journals











IEEE Computer Society

ACM SIGARCH

# Paper: 50 Years History – last section

See what is happening in computer architecture these days and in the future

Multicore, Multithreading, Heterogenous

# Paper: Technology Predictions

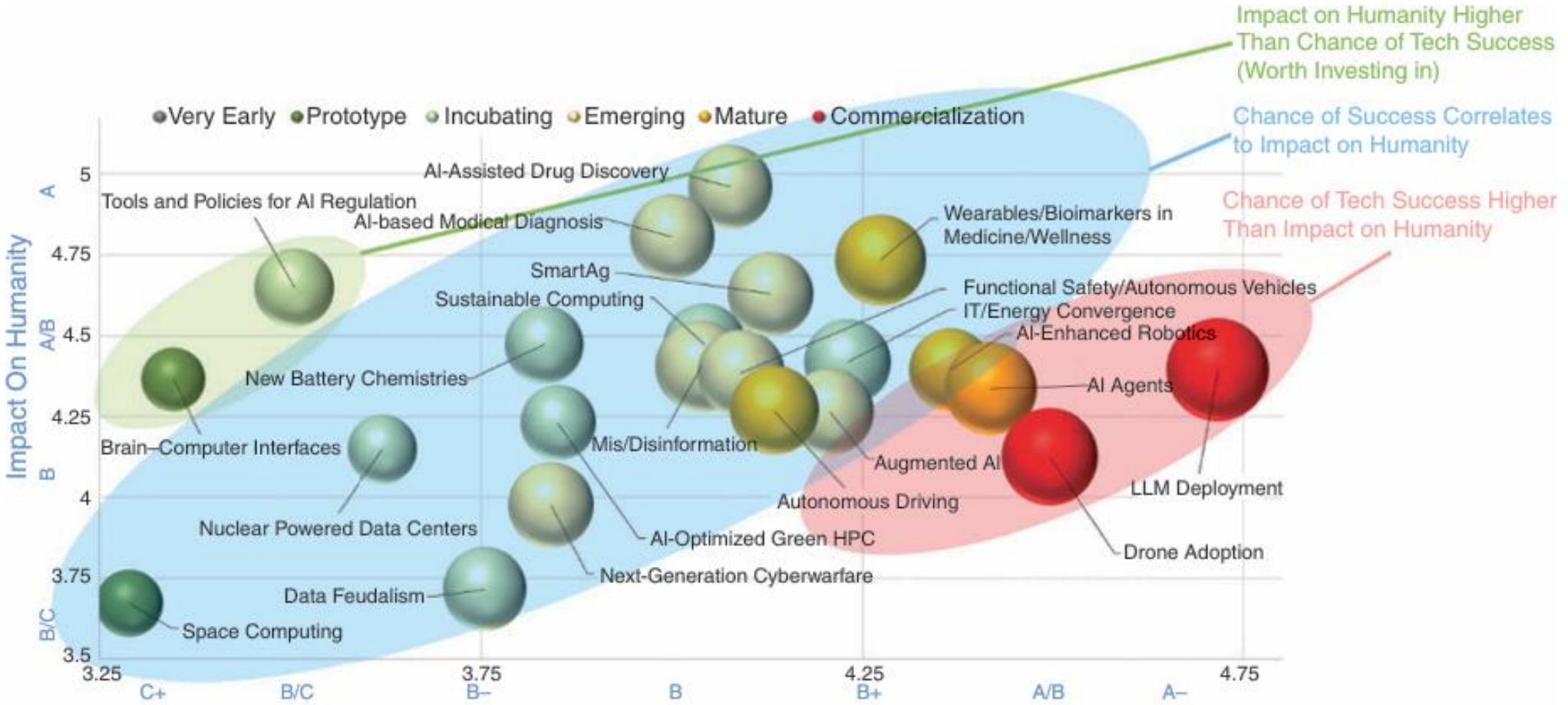



**FIGURE 1.** Comparing 2025 Technology Predictions, Clusters of Correlated Technologies. Technologies highlighted in light green have a higher impact on humanity than the likelihood of success and are worth investing in by governments. Those highlighted in light red have higher chances of success and are worth considering by the industry for further commercialization. Prediction: Tech. development in 2025 (*x*-axis) versus impact to humanity (*y*-axis) (size of bubble proportional to relative market adoption). HPC: high-performance computing; LLM: large language model; SmartAg: smart agriculture.

# Paper: Future … Beyond Moore's Law

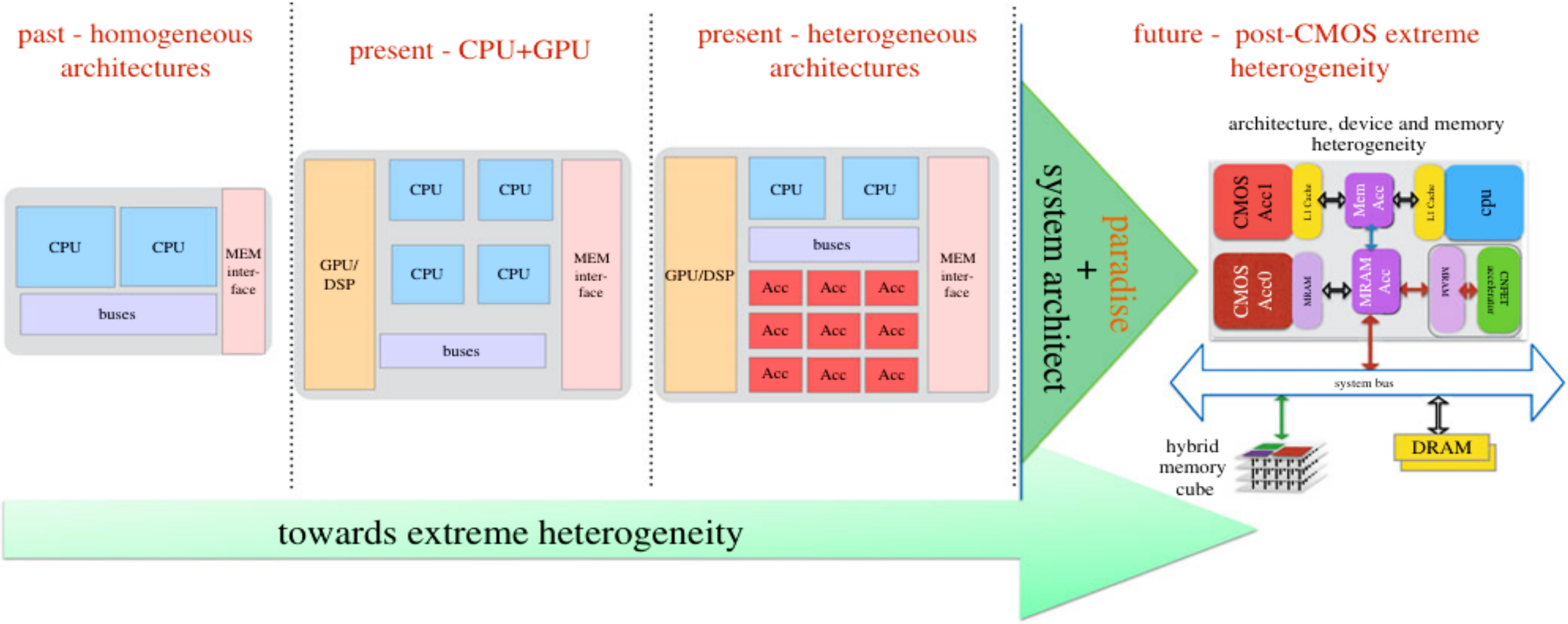
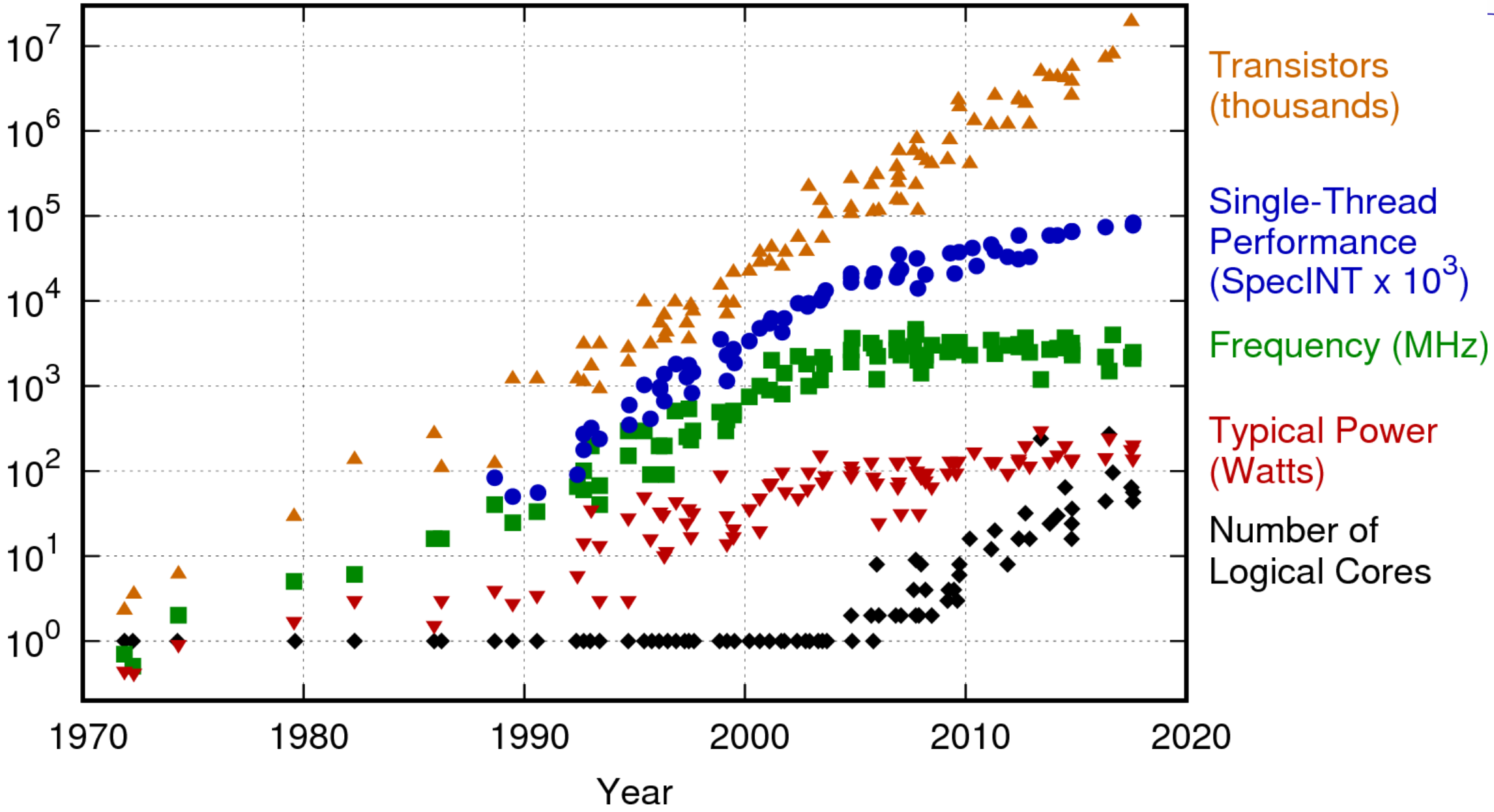


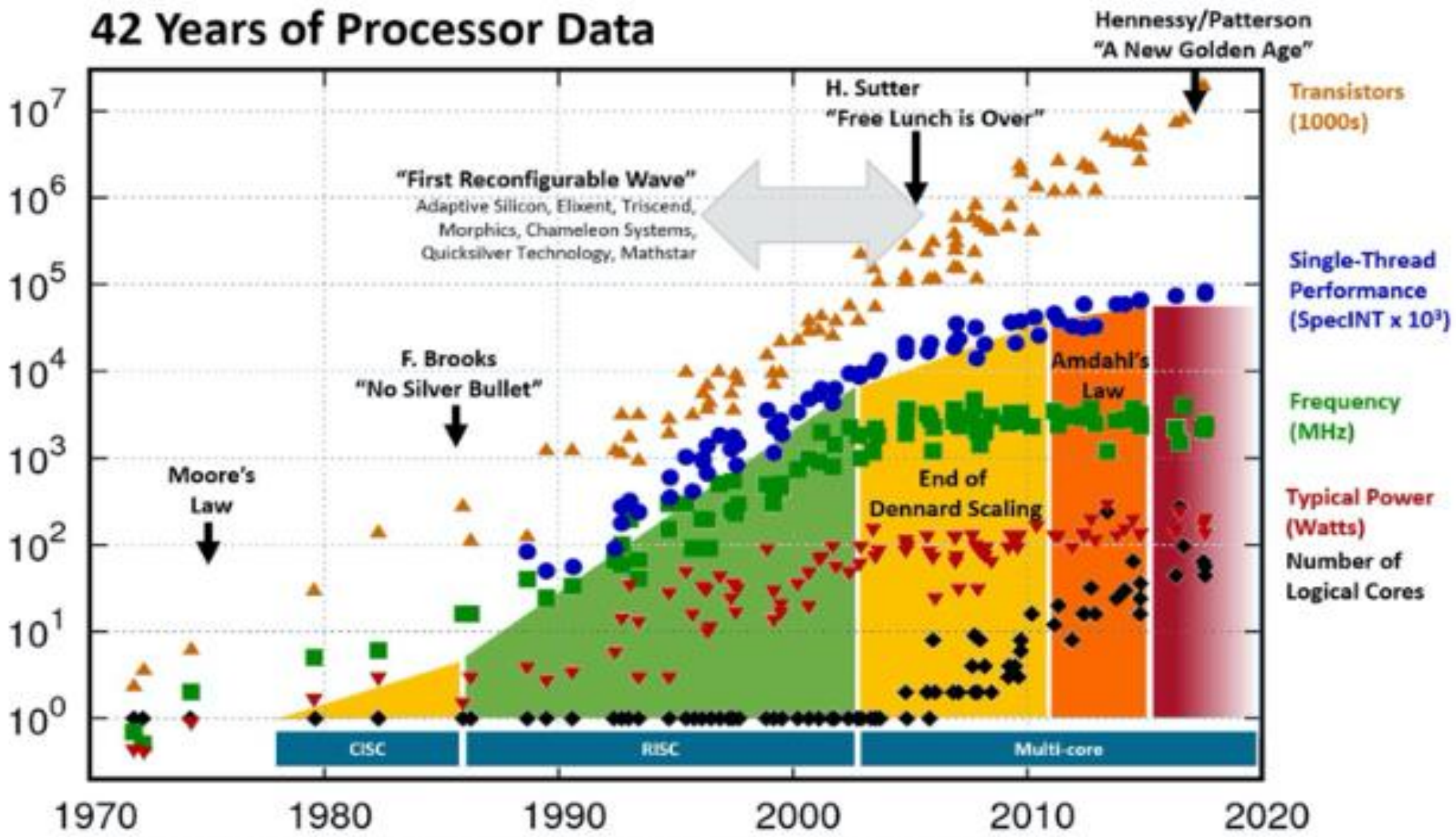

**Figure 4.** Architectural specialization and extreme heterogeneity are anticipated to be the near-term response to the end of classical technology scaling. Figure courtesy of Dilip Vasudevan from LBNL. (Online version in colour.)

# 42 Years of Microprocessor Trend Data

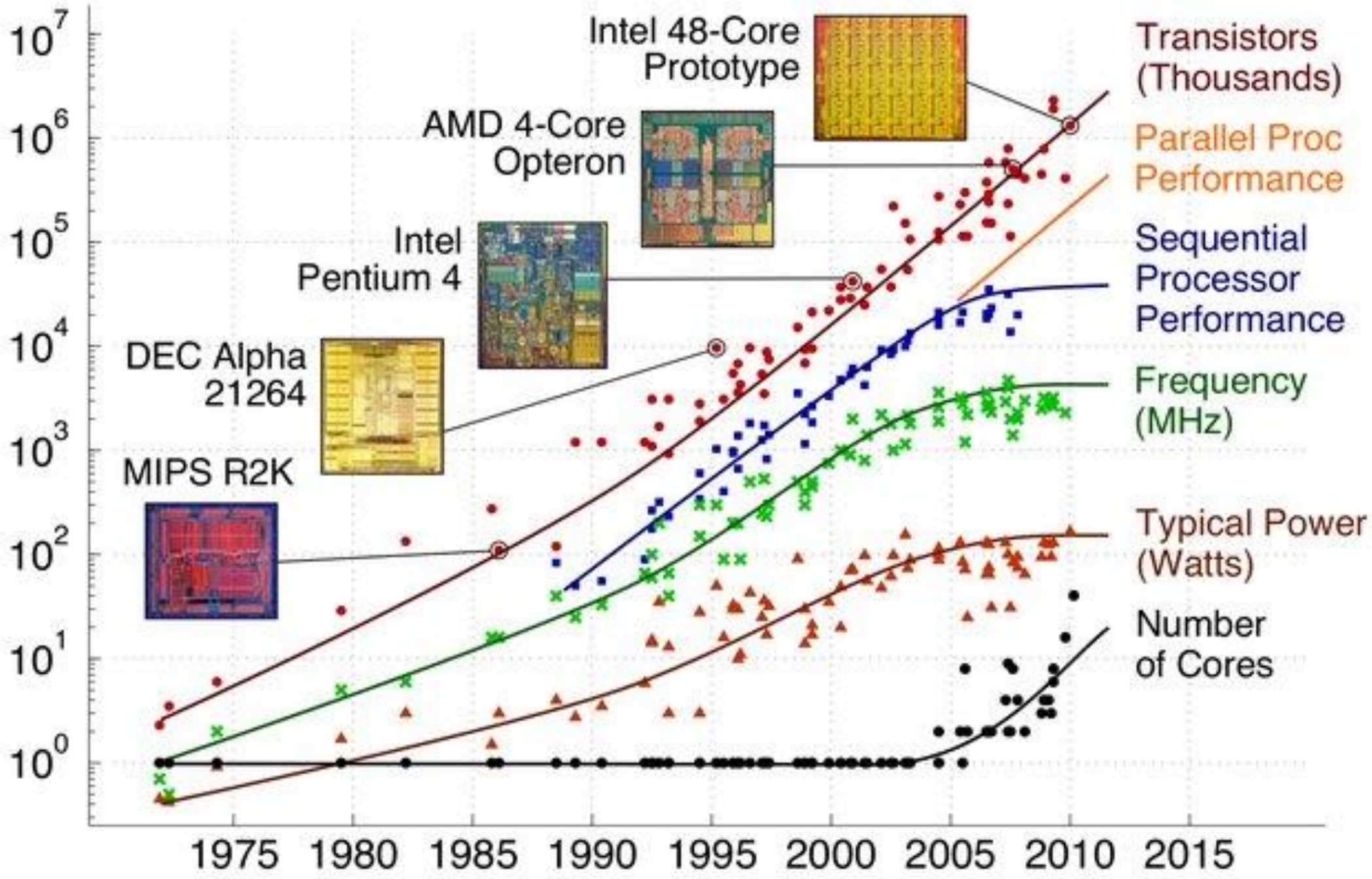



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp

# 42 Years of Processor Data



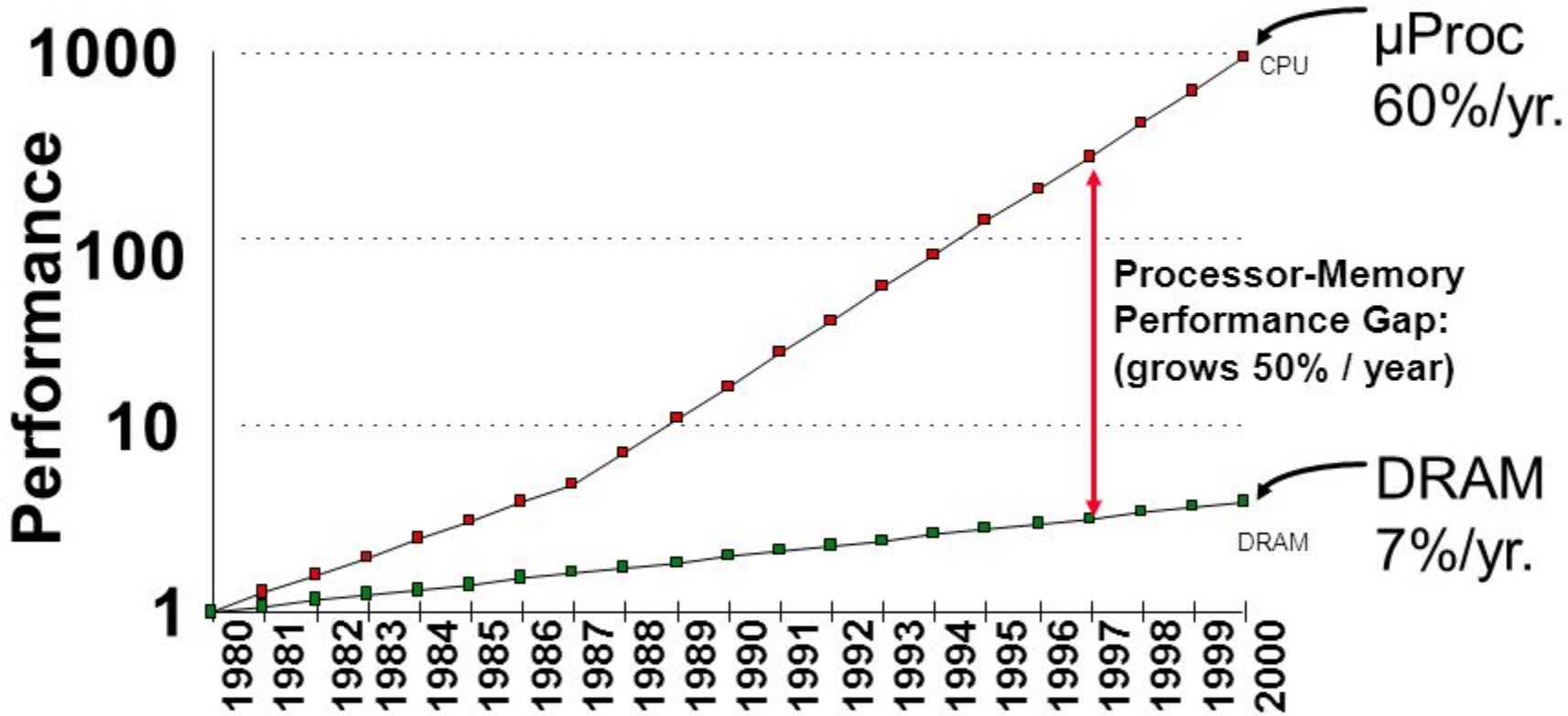

**Hennessy and Patterson, Turing Lecture 2018, overlaid over "42 Years of Processors Data"**
**https://www.karlrupp.net/2018/02/42-years-of-microprocessor-trend-data/;"First Wave" added by Les Wilson, Frank Schirrmeister**
Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp


Intel 48-Core Prototype
AMD 4-Core Opteron
Intel Pentium 4
DEC Alpha 21264
MIPS R2K
Transistors (Thousands)
Parallel Proc Performance
Sequential Processor Performance
Frequency (MHz)
Typical Power (Watts)
Number of Cores
$10^7$
$10^6$
$10^5$
$10^4$
$10^3$
$10^2$
$10^1$
$10^0$
1975 1980 1985 1990 1995 2000 2005 2010 2015

# Memory Hierarchy: Motivation

## Processor-Memory (DRAM) Performance Gap

(i.e. Gap between memory access time and CPU cycle time



# Video of Turing Lecture by Patterson, 2019

https://dl.acm.org/doi/10.1145/3282307

David Patterson - A New Golden Age for Computer Architecture: History, Challenges and Opportunities – YouTube

# Scale in Computing

# The Scale In Computing

| Decimal term | Abbreviation | Value | Binary term | Abbreviation | Value | % Larger |
|---|---|---|---|---|---|---|
| kilobyte | KB | $10^3$ | kibibyte | KiB | $2^{10}$ | 2% |
| megabyte | MB | $10^6$ | mebibyte | MiB | $2^{20}$ | 5% |
| gigabyte | GB | $10^9$ | gibibyte | GiB | $2^{30}$ | 7% |
| terabyte | TB | $10^{12}$ | tebibyte | TiB | $2^{40}$ | 10% |
| petabyte | PB | $10^{15}$ | pebibyte | PiB | $2^{50}$ | 13% |
| exabyte | EB | $10^{18}$ | exbibyte | EiB | $2^{60}$ | 15% |
| zettabyte | ZB | $10^{21}$ | zebibyte | ZiB | $2^{70}$ | 18% |
| yottabyte | YB | $10^{24}$ | yobibyte | YiB | $2^{80}$ | 21% |

**FIGURE 1.1 The $2^x$ vs. $10^y$ bytes ambiguity was resolved by adding a binary notation for all the common size terms.** In the last column we note how much larger the binary term is than its corresponding decimal term, which is compounded as we head down the chart. These prefixes work for bits as well as bytes, so *gigabit* (Gb) is $10^9$ bits while *gibibits* (Gib) is $2^{30}$ bits.

# Approximations in Huge Binary Numbers

- $2^{10}$ is approximately 1 Kilo (approx $1x10^3$)
- $2^{20}$ is approximately 1 Mega (approx $1x10^6$)
- $2^{30}$ is approximately 1 Giga (approx $1x10^9$)

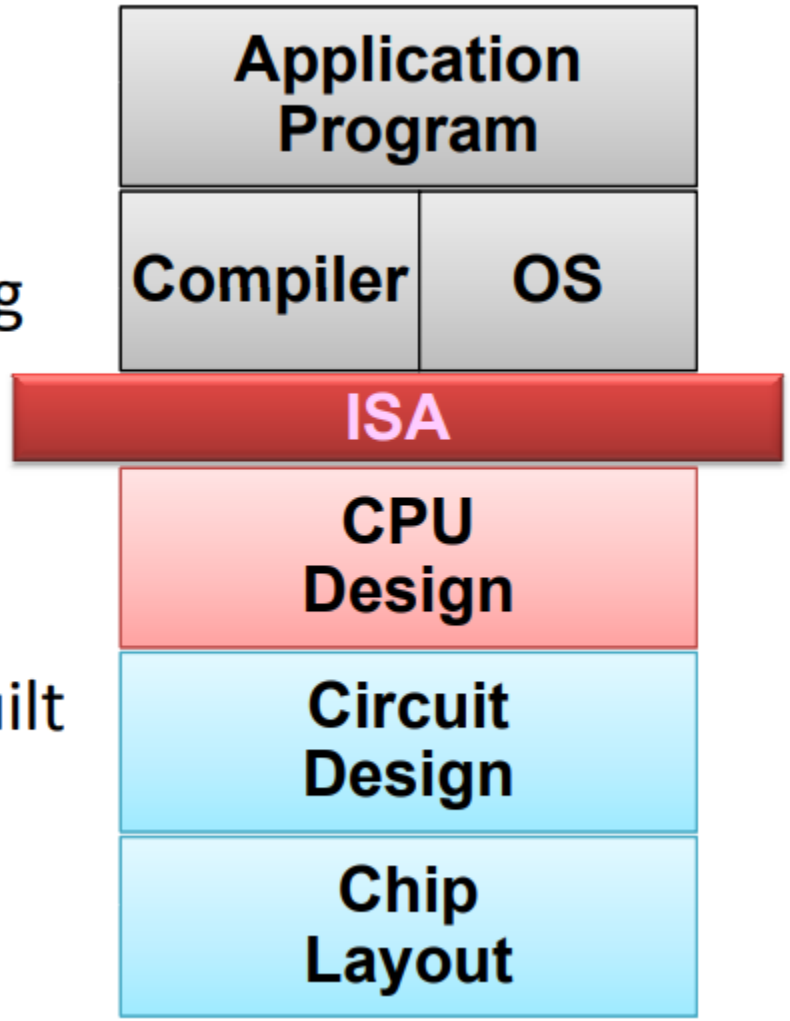
# Introducing Instruction Set Architecture (ISA)

# What is an Instruction Set Architecture?

- Lowest level of Computer Architecture that is visible to a programmer
- ISA comprises instructions in Assembly Language and Corresponding Binary Machine Code
- Primitive syntax compared to a high-level language
- It is easily interpreted and understood by the CPU
- Designed to maximize performance
- Represents features and performance of a CPU
- Designed to minimize cost and design time

# Role of ISA within a Computer

- Assembly Language View
  - Processor state (RF, mem)
  - Instruction set and encoding
- Layer of Abstraction
  - Above: how to program machine - HLL, OS
  - Below: what needs to be built - tricks to make it run fast




**Key:**
**RF = Register File**
**Mem = Memory**
**HLL = High Level Language**
**OS = Operating System**

# RISC-V ISA

## Instruction Set Architecture: The Myopic View of Computer Architecture

We use the term *instruction set architecture* (ISA) to refer to the actual programmer-visible instruction set in this book. The ISA serves as the boundary between the software and hardware. This quick review of ISA will use examples from 80x86, ARMv8, and RISC-V to illustrate the seven dimensions of an ISA. The most popular RISC processors come from ARM (Advanced RISC Machine), which were in 14.8 billion chips shipped in 2015, or roughly 50 times as many chips that shipped with 80x86 processors. Appendices A and K give more details on the three ISAs.

RISC-V ("RISC Five") is a modern RISC instruction set developed at the University of California, Berkeley, which was made free and openly adoptable in response to requests from industry. In addition to a full software stack (compilers, operating systems, and simulators), there are several RISC-V implementations freely available for use in custom chips or in field-programmable gate arrays. Developed 30 years after the first RISC instruction sets, RISC-V inherits its ancestors' good ideas—a large set of registers, easy-to-pipeline instructions, and a lean set of operations—while avoiding their omissions or mistakes. It is a free and open, elegant example of the RISC architectures mentioned earlier, which is why more than 60 companies have joined the RISC-V foundation, including AMD, Google, HP Enterprise, IBM, Microsoft, Nvidia, Qualcomm, Samsung, and Western Digital. We use the integer core ISA of RISC-V as the example ISA in this book.

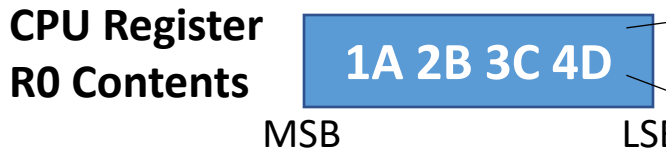
# RISC-V Instructions and Features

- Data Alignment
  - 32-bit bus vs 8-bit memory
  - Big Endian vs Little Endian
- Addressing Modes
- Type and Size of Operands
  - I, R types
- Operations available
  - Arithmetic
  - Logical
  - Shift
- Control Flow Instructions
  - Branch
  - Jump

# Memory Access – Big Endian vs Little Endian

A CPU Register R0 is 32-bits wide.
It has data “1A2B3C4D” Hex.
Store the data in an 8-bit wide RAM that has 4 locations.

Store in Big-Endian style
or Store in Little-Endian Style




| Location | Big Endian RAM |
|---|---|
| 3 | |
| 2 | |
| 1 | |
| 0 | |

| Location | Little Endian RAM |
|---|---|
| 3 | |
| 2 | |
| 1 | |
| 0 | |

# RISC-V Registers and Naming Conventions

| Register | Name | Use | Saver |
|---|---|---|---|
| x0 | zero | The constant value 0 | N.A. |
| x1 | ra | Return address | Caller |
| x2 | sp | Stack pointer | Callee |
| x3 | gp | Global pointer | – |
| x4 | tp | Thread pointer | – |
| x5-x7 | t0-t2 | Temporaries | Caller |
| x8 | s0/fp | Saved register/frame pointer | Callee |
| x9 | s1 | Saved register | Callee |
| x10-x11 | a0-a1 | Function arguments/return values | Caller |
| x12-x17 | a2-a7 | Function arguments | Caller |
| x18-x27 | s2-s11 | Saved registers | Callee |
| x28-x31 | t3-t6 | Temporaries | Caller |
| f0-f7 | ft0-ft7 | FP temporaries | Caller |
| f8-f9 | fs0-fs1 | FP saved registers | Callee |
| f10-f11 | fa0-fa1 | FP function arguments/return values | Caller |
| f12-f17 | fa2-fa7 | FP function arguments | Caller |
| f18-f27 | fs2-fs11 | FP saved registers | Callee |
| f28-f31 | ft8-ft11 | FP temporaries | Caller |

**Figure 1.4 RISC-V registers, names, usage, and calling conventions.** In addition to the 32 general-purpose registers (x0–x31), RISC-V has 32 floating-point registers (f0–f31) that can hold either a 32-bit single-precision number or a 64-bit double-precision number. The registers that are preserved across a procedure call are labeled "Callee" saved.

**RISC-V is a Register-rich Architecture**

# RISC-V Instruction Formats

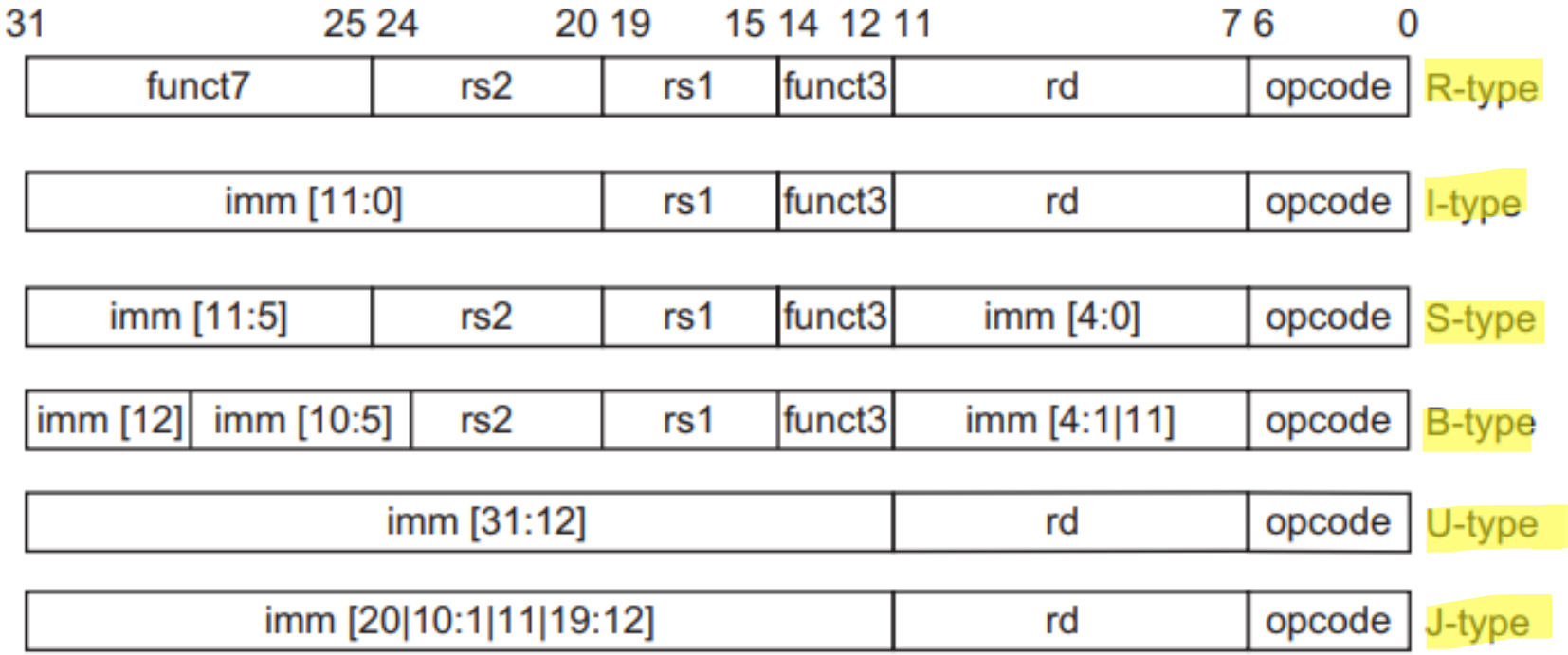



**Figure 1.7 The base RISC-V instruction set architecture formats.** All instructions are 32 bits long. The R format is for integer register-to-register operations, such as ADD, SUB, and so on. The I format is for loads and immediate operations, such as LD and ADDI. The B format is for branches and the J format is for jumps and link. The S format is for stores. Having a separate format for stores allows the three register specifiers (rd, rs1, rs2) to always be in the same location in all formats. The U format is for the wide immediate instructions (LUI, AUIPC).

# RISC-V Data type Instructions

| Instruction type/opcode | Instruction meaning |
|---|---|
| *Data transfers* | *Move data between registers and memory, or between the integer and FP or special registers; only memory address mode is 12-bit displacement+contents of a GPR* |
| `lb, lbu, sb` | Load byte, load byte unsigned, store byte (to/from integer registers) |
| `lh, lhu, sh` | Load half word, load half word unsigned, store half word (to/from integer registers) |
| `lw, lwu, sw` | Load word, load word unsigned, store word (to/from integer registers) |
| `ld, sd` | Load double word, store double word (to/from integer registers) |
| `flw, fld, fsw, fsd` | Load SP float, load DP float, store SP float, store DP float |
| `fmv._.x, fmv.x._` | Copy from/to integer register to/from floating-point register; "__"=S for single-precision, D for double-precision |
| `csrrw, csrrwi, csrrs, csrrsi, csrrc, csrrci` | Read counters and write status registers, which include counters: clock cycles, time, instructions retired |

# RISC-V Arithmetic / Logical Instructions

| *Arithmetic/logical* | *Operations on integer or logical data in GPRs* |
|---|---|
| `add, addi, addw, addiw` | Add, add immediate (all immediates are 12 bits), add 32-bits only & sign-extend to 64 bits, add immediate 32-bits only |
| `sub, subw` | Subtract, subtract 32-bits only |
| `mul, mulw, mulh, mulhsu, mulhu` | Multiply, multiply 32-bits only, multiply upper half, multiply upper half signed-unsigned, multiply upper half unsigned |
| `div, divu, rem, remu` | Divide, divide unsigned, remainder, remainder unsigned |
| `divw, divuw, remw, remuw` | Divide and remainder: as previously, but divide only lower 32-bits, producing 32-bit sign-extended result |
| `and, andi` | And, and immediate |
| `or, ori, xor, xori` | Or, or immediate, exclusive or, exclusive or immediate |
| `lui` | Load upper immediate; loads bits 31-12 of register with immediate, then sign-extends |
| `auipc` | Adds immediate in bits 31–12 with zeros in lower bits to PC; used with `JALR` to transfer control to any 32-bit address |
| `sll, slli, srl, srli, sra, srai` | Shifts: shift left logical, right logical, right arithmetic; both variable and immediate forms |
| `sllw, slliw, srlw, srliw, sraw, sraiw` | Shifts: as previously, but shift lower 32-bits, producing 32-bit sign-extended result |
| `slt, slti, sltu, sltiu` | Set less than, set less than immediate, signed and unsigned |

# RISC-V Control Instructions

| *Control* | *Conditional branches and jumps; PC-relative or through register* |
|---|---|
| `beq`, `bne`, `blt`, `bge`, `bltu`, `bgeu` | Branch GPR equal/not equal; less than; greater than or equal, signed and unsigned |
| `jal`, `jalr` | Jump and link: save PC+4, target is PC-relative (`JAL`) or a register (`JALR`); if specify `x0` as destination register, then acts as a simple jump |
| `ecall` | Make a request to the supporting execution environment, which is usually an OS |
| `ebreak` | Debuggers used to cause control to be transferred back to a debugging environment |
| `fence`, `fence.i` | Synchronize threads to guarantee ordering of memory accesses; synchronize instructions and data for stores to instruction memory |

# Performance of Computer Systems

# Simple Computer Performance Measures

- Time to execute a program – in seconds
- Clock Speed of Computer – Mega Hertz or Giga Hz
- Comparing Benchmarks
- In terms of the width of Data Bus (16 bit, 32 bit, 64 bit)
- In terms of the size of different memory (8GB / 256GB)
  - Cache size
  - RAM
  - Hard disk
  - SSD, etc.
- In terms of multiple cores and multiple I/O available
- Software
  - Algorithm
  - Language / Compiler
  - Optimization – Parallel, Vector, etc.

**In terms of Resources**

# What is MFLOPS or Giga Flops or Tera Flops?

FLOPS = Floating Point Linear Operations Per Second

$$\text{MFLOPS } rate = \frac{Number\ of\ executed\ floating\ point\ operations\ in\ a\ program}{Execution\ time\ \times 10^6}$$

# Principles of Modern Computer Design

- Take Advantage of Parallelism
  - e.g. multiple processors, disks, memory banks, pipelining, multiple functional units
- Principle of Locality
  - Reuse of data and instructions (i.e. in Caches)
- Focus on the Common Case
  - Amdahl's Law

$$Speedup_{overall} = \frac{1}{(1 - Fraction_{enhanced}) + \frac{Fraction_{enhanced}}{Speedup_{enhanced}}}$$

# Energy and Power

# CPU Architecture Today

## Heat becoming un unmanageable problem

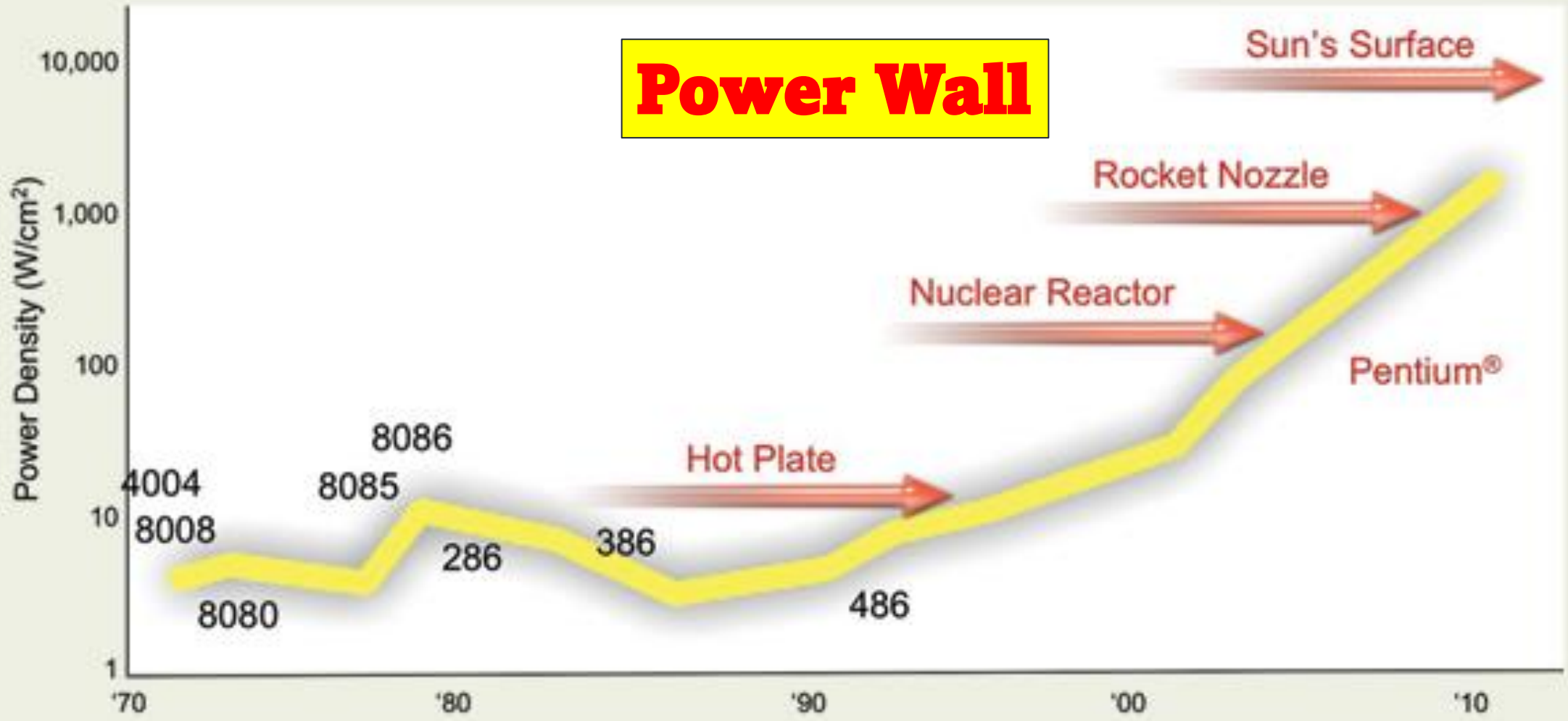



**Figure 1.** In CPU architecture today, heat is becoming an unmanageable problem. (Courtesy of Pat Gelsinger, Intel Developer Forum, Spring 2004)

# Energy and Power in a Microprocessor

## Energy and Power Within a Microprocessor

For CMOS chips, the traditional primary energy consumption has been in switching transistors, also called *dynamic energy*. The energy required per transistor is proportional to the product of the capacitive load driven by the transistor and the square of the voltage:

$$\text{Energy}_{\text{dynamic}} \propto \text{Capacitive load} \times \text{Voltage}^2$$

This equation is the energy of pulse of the logic transition of $0 \rightarrow 1 \rightarrow 0$ or $1 \rightarrow 0 \rightarrow 1$. The energy of a single transition ($0 \rightarrow 1$ or $1 \rightarrow 0$) is then:

$$\text{Energy}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2$$

The power required per transistor is just the product of the energy of a transition multiplied by the frequency of transitions:

$$\text{Power}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency switched}$$

For a fixed task, slowing clock rate reduces power, but not energy.

Clearly, dynamic power and energy are greatly reduced by lowering the voltage, so voltages have dropped from 5 V to just under 1 V in 20 years. The capacitive load is a function of the number of transistors connected to an output and the technology, which determines the capacitance of the wires and the transistors.

# Example of Energy calculation

**Example** Some microprocessors today are designed to have adjustable voltage, so a 15% reduction in voltage may result in a 15% reduction in frequency. What would be the impact on dynamic energy and on dynamic power?

***Answer*** Because the capacitance is unchanged, the answer for energy is the ratio of the voltages

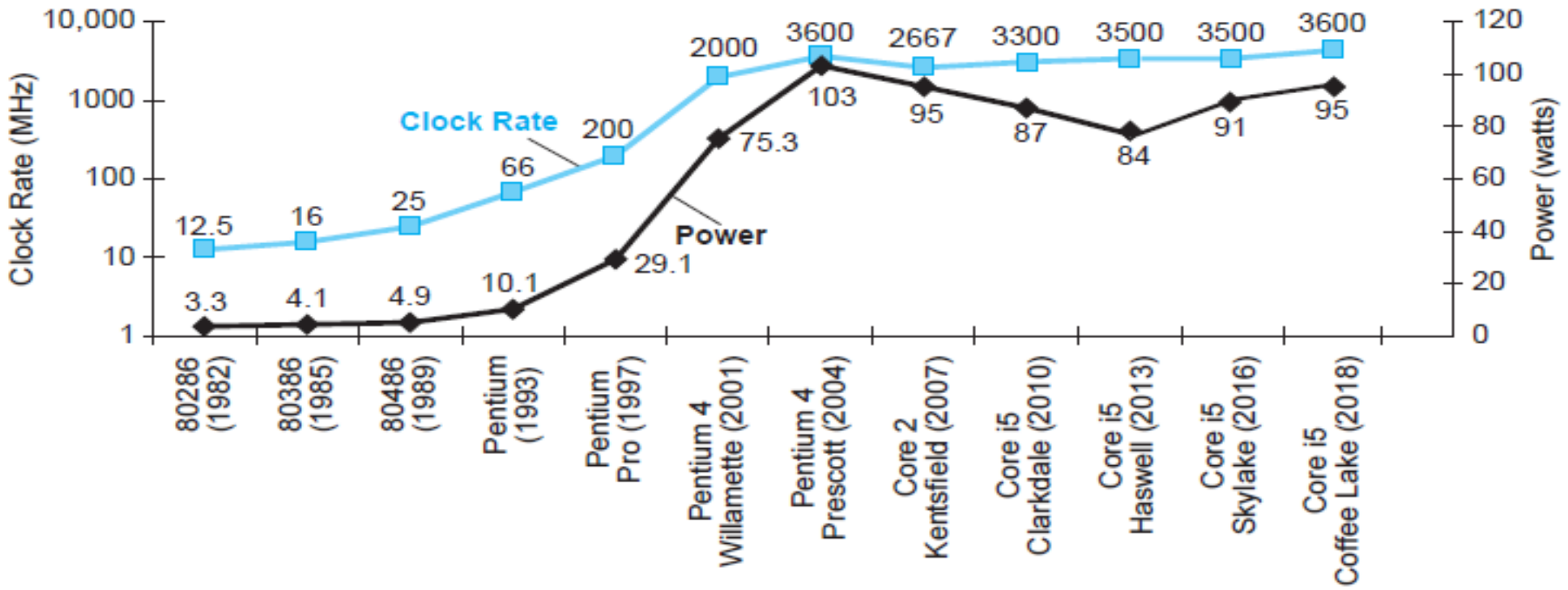
$$\frac{\text{Energy}_{\text{new}}}{\text{Energy}_{\text{old}}} = \frac{(\text{Voltage} \times 0.85)^2}{\text{Voltage}^2} = 0.85^2 = 0.72$$

which reduces energy to about 72% of the original. For power, we add the ratio of the frequencies

$$\frac{\text{Power}_{\text{new}}}{\text{Power}_{\text{old}}} = 0.72 \times \frac{(\text{Frequency switched} \times 0.85)}{\text{Frequency switched}} = 0.61$$

shrinking power to about 61% of the original.

# CPU Power Trends




- In CMOS IC technology

$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

×30 | 5V → 1V | ×1000

# Readings

- H&P Computer Architecture Textbook chapter 1
- Research Papers for new Computing architectures and paradigm
  - Challenging Trends in Energy of Computing for Data Centers, IEEE Computer, 2024
  - The 50 Year History of the Microprocessor as Five Technology Eras, IEEE Micro, 2021
  - A New Golden Age for Computer Architecture, Turing Lecture, Communications of the ACM, 2019
- See end of chapter questions from H&P textbook